

# Multi-Objective Optimization

## Main Classes of Multi-Objective Genetic Algorithms: Core Ideas and Mechanisms

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# Lecture Overview

## Main Classes of Multi-Objective Genetic Algorithms

### 1 Foundations

- Why scalarization is not enough
- Pareto dominance and non-dominated solutions

### 2 **VEGA** — Vector Evaluated Genetic Algorithm (Schaffer, 1985)

### 3 **MOGA** — Multi-Objective Genetic Algorithm (Murata & Ishibuchi, 1995)

### 4 **NSGA** — Non-dominated Sorting Genetic Algorithm (Srinivas & Deb, 1994)

### 5 **SPEA** — Strength Pareto Evolutionary Algorithm (Zitzler & Thiele, 1999)

### 6 **PESA** — Pareto Envelope-based Selection Algorithm (Corne et al., 2000)

### 7 Comparison and lessons learned

# Why Multi-Objective GAs?

## Foundations

### The Problem with Weighted Sums:

- Scalarization fixes a search direction in objective space
- Cannot find concave Pareto fronts
- Requires *a priori* weight knowledge

### Pareto Concepts:

- **Dominance:**  $a \succ b$  iff  $a$  is no worse in all objectives and better in at least one
- **Non-dominated set:** solutions not dominated by any other
- **Pareto-optimal front:** image of the non-dominated set in objective space

### Goal of a Multi-Objective GA:

- Find a **set** of solutions approximating the Pareto front
- Achieve both **convergence** (close to front) and **diversity** (well spread)

## Historical Context:

- First multi-objective GA extension
- Motivated by problems with non-commensurable objectives

## Core Idea:

- Selection is performed **separately for each objective**
- Population divided into subpopulations, each governed by one objective
- Shuffle and recombine to produce next generation

## Selection Mechanism:

- For  $M$  objectives, the mating pool is split into  $M$  equal parts
- Part  $i$  filled with individuals selected according to objective  $f_i$
- All parts shuffled together, then crossover and mutation applied

### Strengths:

- Simple and intuitive extension of single-objective GA
- Protects the best individuals on each dimension
- Provides appropriate probabilities for multiple selection of good all-round individuals
- Has a built-in advantage: works with a population of test solutions

### Limitations:

- Biased toward extreme solutions; search directions are parallel to axes
- Cannot find middle (trade-off) Pareto-optimal solutions
- Subpopulations tend to converge to individual optima

### Schaffer's Suggested Improvements:

- Heuristic preference for non-dominated individuals
- Mate selection to crossbreed among species

## Motivation:

- Overcome VEGA's bias toward extreme solutions
- VEGA's search directions are fixed and parallel to axes

## Core Idea — Random Weighted Sum:

- Combine multiple objectives into a scalar fitness via weighted sum
- **Weights are not constant** — randomly specified for each selection
- Produces **various search directions** in objective space

## Weight Specification:

$$w_i = \frac{\text{random}_i}{\sum_{j=1}^n \text{random}_j}, \quad i = 1, 2, \dots, n$$

- $\text{random}_j$  is a non-negative random number
- Each pair of strings is selected with different weights

### Elite Preserve Strategy:

- Maintain a tentative set of Pareto-optimal solutions
- Updated at every generation
- $N_{\text{elite}}$  individuals randomly selected from this set
- Inherited to next generation as elite individuals
- Effect: keeps variety in each population

### Algorithm Outline:

- 1 **Init.:** generate  $N_{\text{pop}}$  strings
- 2 **Evaluation:** compute objectives; update tentative Pareto set
- 3 **Selection:** fitness with random weights; select  $N_{\text{pop}}/2$  pairs
- 4 **Crossover:** generate  $N_{\text{pop}}$  new strings
- 5 **Mutation:** apply with prespecified probability
- 6 **Elitism:** replace  $N_{\text{elite}}$  random strings with elites
- 7 **Termination:** check stopping condition; else go to Step 1

### Strengths:

- Can handle **concave Pareto fronts** (unlike fixed weighted sum)
- Various search directions — finds solutions along the whole front
- Elite preservation maintains diversity
- Found all Pareto-optimal solutions on the “unitation versus pairs” test problem

### Limitations:

- Random weights may not uniformly cover all Pareto regions
- Still relies on scalarization at each selection step
- Concave fronts handled, but distribution may be uneven

# NSGA

Non-dominated Sorting Genetic Algorithm (Srinivas & Deb, 1994)

## Motivation:

- VEGA suffers from bias toward some Pareto-optimal solutions
- Goldberg's suggestion: use non-dominated sorting + niching

## Core Idea — Nondominated Sorting:

- 1 Identify non-dominated individuals in current population
- 2 Assign them a large **dummy fitness** value
- 3 Share this fitness using **sharing function**
- 4 Temporarily ignore them; process rest of population to identify second front
- 5 Assign smaller dummy fitness; repeat until all classified

## Key Mechanisms:

- **Ranking selection:** emphasizes good points
- **Niche method:** maintains stable subpopulations of good points
- **Sharing:** degrades fitness by number of individuals nearby

### Sharing Function:

$$Sh(d_{ij}) = \begin{cases} 1 - \left( \frac{d_{ij}}{\sigma_{share}} \right)^2, & d_{ij} < \sigma_{share}, \\ 0, & \text{otherwise.} \end{cases}$$

- $d_{ij}$ : phenotypic distance between  $i$  and  $j$
- $\sigma_{share}$ : max. distance to be in the same niche
- Shared fitness = dummy fitness / niche count

### Algorithm Flow:

- 1 Rank population by non-domination
- 2 Identify first front; assign dummy fitness
- 3 Share fitness; ignore front
- 4 Repeat for second, third front, etc.
- 5 Reproduce according to shared dummy fitness
- 6 Stochastic remainder proportionate selection

**Algorithm.** NSGA (Srinivas & Deb, 1994)

**Require:**  $N$  (pop. size),  $G_{\max}$ ,  $p_c$ ,  $p_m$ ,  $\sigma_{\text{share}}$

- 1: Generate initial population  $P(0)$  of size  $N$ ; evaluate objectives;  $t \leftarrow 0$
- 2: **while**  $t < G_{\max}$  **do**
- 3:   Rank by non-dominated sorting: identify front  $F_1$ , assign large dummy fitness
- 4:   Share fitness within  $F_1$ :  $Sh(d_{ij}) = 1 - (d_{ij}/\sigma_{\text{share}})^2$  if  $d_{ij} < \sigma_{\text{share}}$ , else 0;  
     $\text{shared}_i = \text{dummy}_i / \sum_j Sh(d_{ij})$
- 5:   Temporarily ignore  $F_1$ ; repeat for  $F_2, F_3, \dots$  with smaller dummy fitness until all ranked
- 6:   Select parents by stochastic remainder proportionate selection
- 7:   Apply crossover ( $p_c$ ) and mutation ( $p_m$ ) to generate  $P(t+1)$
- 8:   Evaluate offspring;  $t \leftarrow t + 1$
- 9: **end while**
- 10: **return** non-dominated solutions in  $P(t)$

### Strengths:

- Eliminates VEGA's bias — covers the entire Pareto front
- Stable, uniform reproductive potential across non-dominated individuals
- Handles any number of objectives
- Works for min. and max. problems

### Limitations:

- Computationally expensive
- No elitism in original form
- $\sigma_{\text{share}}$  needs tuning
- Later addressed by NSGA-II

### Results (problem F1):

- NSGA kept  $\approx 10$  individuals per subregion over 500 generations
- VEGA populated at most 3 subregions (near individual optima)
- NSGA distributed successfully over disjointed Pareto fronts

# SPEA

Strength Pareto Evolutionary Algorithm (Zitzler & Thiele, 1999)

## Motivation:

- Combine strengths of previous MOEAs in a unique manner
- Introduce elitism via external population
- No distance parameter needed for niching

## Unique Features:

- 1 **External archive** storing non-dominated solutions
- 2 Fitness based on **strength** — number of dominated population members
- 3 **Pareto-based niching** — no niche radius required
- 4 **Clustering** to reduce archive without destroying characteristics

## Two Populations:

- $P$ : regular population
- $P'$ : external non-dominated set
- Both participate in selection

### Two-Stage Fitness Assignment:

#### Stage 1 — External Pop. $P'$ :

- Solution  $i \in P'$  assigned strength  $s_i$
- $s_i \propto$  dominated members
- $s_i = \frac{n}{N+1}$  ( $n$  = dominated members,  $N$  = population size)
- Fitness of  $i$ :  $f_i = s_i$

#### Stage 2 — Population $P$ :

- Fitness of  $j$  = sum of strengths of external solutions covering  $j$
- $f_j = 1 + \sum_{i \succeq j} s_i \in [1, N)$
- Members of  $P'$  are fitter than  $P$

### Intuition:

- “Stronger” non-dominated solution  $\rightarrow$  less “fit” covered individuals
- Prefers individuals near Pareto front and distributes them along the tradeoff surface

**Algorithm.** SPEA (Zitzler & Thiele, 1999)

**Require:**  $N$  (pop. size),  $N'$  (archive size),  $G_{\max}$

- 1: Generate  $P(0)$  of size  $N$ ; set archive  $P'(0) \leftarrow \emptyset$ ;  $t \leftarrow 0$
- 2: **while**  $t < G_{\max}$  **do**
- 3:   Copy non-dominated individuals from  $P(t)$  into  $P'(t)$ ; remove dominated solutions from  $P'(t)$
- 4:   **if**  $|P'(t)| > N'$  **then** apply average linkage clustering to reduce  $P'(t)$  to size  $N'$
- 5:   **end if**
- 6:   Evaluate fitness: for  $i \in P'(t)$ ,  $s_i \leftarrow |\{j \in P(t) : i \succeq j\}| / (N + 1)$ ,  $f_i \leftarrow s_i$ ; for  $j \in P(t)$ ,  
 $f_j \leftarrow 1 + \sum_{i \succeq j, i \in P'(t)} s_i$
- 7:   Select parents from  $P(t) \cup P'(t)$  via binary tournament
- 8:   Apply crossover and mutation to generate offspring  $P(t + 1)$
- 9:    $P'(t + 1) \leftarrow P'(t)$ ;  $t \leftarrow t + 1$
- 10: **end while**
- 11: **return** archive  $P'(t)$

### Why Clustering?

- Pareto set may be extremely large or infinite
- Too many non-dominated solutions reduce selection pressure
- Non-uniform distribution biases fitness assignment

### Average Linkage Clustering:

- 1 Each external point starts as its own distinct cluster
- 2 While  $|C| > N'$ : find the two closest clusters and merge them
- 3 Select a representative (centroid) for each remaining cluster

### Effect:

- Archive  $P'$  is truncated to a fixed size  $N'$
- Reduces redundancy while preserving the overall shape of the front (see full flow in the pseudocode)

### Strengths:

- **Elitism is crucial** for performance
- No niche radius parameter needed
- Can find solutions closer to Pareto front than single-objective EA
- Outperformed four other MOEAs on 0/1 knapsack problem
- Covers 100% and dominates 50% of RTS+Pareto ranking solutions on system synthesis

### Limitations:

- Archive size  $N'$  influences niching capability
- Clustering may remove important extreme solutions
- Strength-based fitness can be complex to tune

### Overview (Corne et al., 2000):

- Evolutionary algorithm designed to integrate selection and diversity maintenance
- Uses the **same core technique** for both tasks
- Simple structure to describe and implement

### Two-Population Model:

- **Internal Population ( $P_I$ ):** stores new candidate solutions generated during search
- **External Population ( $P_E$ ):** fixed-capacity archive of non-dominated solutions

### Core Concept:

- An implicit hyper-grid divides the phenotype space into discrete hyper-boxes

### Squeeze Factor:

- Defined for each chromosome in the external population
- **Squeeze Factor** = number of other chromosomes sharing the same hyper-box
- Quantifies crowd density in specific regions of the Pareto front

### Role in PESA:

- Directs both the **selection of parents** and the **archive maintenance**

**Algorithm.** PESA (Corne et al., 2000)

**Require:**  $N$  (internal pop. size),  $A_{\max}$  (archive size),  $G_{\max}$ ,  $p_c$

- 1: Generate  $P_I(0)$  of size  $N$ ; evaluate objectives;  $P_E(0) \leftarrow \emptyset$
- 2: Incorporate non-dominated members of  $P_I(0)$  into  $P_E(0)$ ;  $t \leftarrow 0$
- 3: **while**  $t < G_{\max}$  **do**
- 4:     Clear  $P_I(t)$  and generate  $N$  candidates: w.p.  $p_c$ , tournament on squeeze factor (lowest wins) for two parents  $\rightarrow$  crossover + mutation; w.p.  $1 - p_c$ , tournament for one parent  $\rightarrow$  mutation only
- 5:     Evaluate objectives of  $P_I(t)$
- 6:     Update archive: add non-dominated candidates from  $P_I(t)$ , remove dominated ones; if  $|P_E(t)| > A_{\max}$ , remove member with highest squeeze factor
- 7:      $t \leftarrow t + 1$
- 8: **end while**
- 9: **return** archive  $P_E(t)$

### Strengths:

- **Unified approach** — same crowding measure for selection and diversity
- Simple to describe and implement
- No distance parameter like  $\sigma_{share}$
- Performs well at all time limits (1k, 5k, 20k evaluations)
- Overall best on Deb's T1–T6

### Limitations:

- Hyper-grid size needs tuning ( $32 \times 32$  used)
- Squeeze factor may not capture all diversity aspects
- Archive size  $P_E$  influences performance

### Results Summary (T1–T6):

- Best or joint best on 5 of 6 test functions at 20k evaluations
- Never worst of the three algorithms (PESA, SPEA, PAES)
- Best on easy T1 (hill-climbing suitable)
- SPEA best on deceptive T5

# Comparison of the Five Main MOGA Classes

## Summary Table

| Algorithm | Selection Basis            | Diversity Mechanism                         | Elitism                |
|-----------|----------------------------|---------------------------------------------|------------------------|
| VEGA      | Separate per objective     | Implicit via subpopulations                 | No                     |
| MOGA      | Random weighted sum        | Random weights + elite preserve             | Yes (tentative Pareto) |
| NSGA      | Non-dominated sorting      | Fitness sharing ( $\sigma_{\text{share}}$ ) | No                     |
| SPEA      | Strength (dominated count) | Pareto-based niching + clustering           | Yes (external archive) |
| PESA      | Squeeze factor             | Hyper-grid crowding                         | Yes (external archive) |

### Key Evolution:

- VEGA → MOGA: random weights overcome extreme bias
- MOGA → NSGA: explicit non-dominated sorting
- NSGA → SPEA: elitism + parameter-free niching
- SPEA → PESA: unified selection and diversity

# Lessons Learned

## Key Takeaways

- ❶ **Scalarization alone is insufficient**
  - Fixed weights cannot find concave fronts
  - Random weights (MOGA) help but may not cover all regions
- ❷ **Pareto dominance is central**
  - NSGA, SPEA, PESA all explicitly use dominance
  - Enables parameter-free niching (SPEA) and unified selection (PESA)
- ❸ **Elitism matters**
  - SPEA and PESA with archives outperform non-elitist NSGA
  - External population preserves good solutions
- ❹ **Diversity maintenance is essential**
  - Sharing (NSGA), strength niching (SPEA), hyper-grid (PESA)
  - Without it, population converges to a few extreme points
- ❺ **No single best algorithm**
  - PESA best overall on Deb's functions
  - Choice depends on problem characteristics

# Key References

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